SQL Basics Lab - Student Workbook

Lab OverviewIn this lab, you will practice SQL SELECT and WHERE statements using a company database. The database contains information about employees, products, and orders. Work through each exercise sequentially and write your SQL queries to solve the problems.

Database Schema

You will work with three tables:

employees

- employee_id (INT, Primary Key)
- first_name (VARCHAR)
- last_name (VARCHAR)
- email (VARCHAR)
- department (VARCHAR)
- job_title (VARCHAR)
- salary (DECIMAL)
- hire_date (DATE)
- manager_id (INT)
- city (VARCHAR)

products

- product_id (INT, Primary Key)
- product_name (VARCHAR)
- category (VARCHAR)
- price (DECIMAL)
- stock_quantity (INT)
- supplier (VARCHAR)

orders

- order_id (INT, Primary Key)
- customer_name (VARCHAR)
- product_id (INT)
- quantity (INT)
- order_date (DATE)
- total_amount (DECIMAL)
- status (VARCHAR)

Instructions

1. Read each exercise carefully

2. Write your SQL query below each exercise

3. Test your query to ensure it returns the expected results

4. Verify the number of rows returned matches the expected result

5. Ask for help if you're stuck

Part 1: Basic SELECT Queries

Exercise 1.1: Select All Columns

Task: Retrieve all information from the employees table.

Expected Result: 15 rows with all columns

Your Query:

SELECT * FROM employees;

Output

```
7 •  SELECT *
8    FROM employees;
9
```

ult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Task: Retrieve only the first name, last name, and email of all employees.

Exercise 1.2: Select Specific Columns

Expected Result: 15 rows with 3 columns

Your Query:

SELECT first_name, last_name, email

FROM employees;

Output

```
79 •  SELECT first_name, last_name, email
80    FROM employees;
81
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	email
John	Watson	john.watson@company.com
Jane	Wilson	jane.wilson@company.com
Mark	Johnson	mark.johnson@company.com
Sara	Anderson	sara.anderson@company.com
Paul	Brown	paul.brown@company.com
Emily	Clark	emily.clark@company.com
David	Miller	david.miller@company.com
Anna	Taylor	anna.taylor@company.com
Robert	Wilson	robert.wilson@company.com
James	Thomas	james.thomas@company.com
Linda	Martinez	linda.martinez@gmail.com
Kevin	White	kevin.white@gmail.com
Rachel	Green	rachel.green@gmail.com
Brian	Adams	brian.adams@gmail.com
Nancy	Scott	nancy.scott@gmail.com

Exercise 1.3: Using Column Aliases

Task: Retrieve employee first name as "First Name", last name as "Last Name", and salary as "Annual Salary".

Expected Result: 15 rows with renamed column headers

Your Query:

```
SELECT
   first_name AS "First Name",
   last_name AS "Last Name",
   salary AS "Annual Salary"
FROM employees;
```

Output

```
SELECT
    first_name AS "First Name",
    last_name AS "Last Name",
    salary AS "Annual Salary"
FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

First Name	Last Name	Annual Salary
John	Watson	90000.00
Jane	Wilson	70000.00
Mark	Johnson	55000.00
Sara	Anderson	65000.00
Paul	Brown	85000.00
Emily	Clark	75000.00
David	Miller	60000.00
Anna	Taylor	78000.00
Robert	Wilson	52000.00
James	Thomas	54000.00
Linda	Martinez	72000.00
Kevin	White	58000.00
Rachel	Green	60000.00
Brian	Adams	68000.00
Nancy	Scott	56000.00

Exercise 1.4: SELECT DISTINCT

Task: Find all unique departments in the company.

Expected Result: 5 unique departments

Your Query:

SELECT DISTINCT department

FROM employees;

Output

```
SELECT DISTINCT department
FROM employees;
```

Result Grid | Filter Rows:

department
IT
HR
Sales
Marketing
Finance

Exercise 1.5: SELECT DISTINCT with Multiple Columns

Task: Find all unique combinations of department and city.

Expected Result: Multiple unique combinations

Your Query:

```
SELECT DISTINCT department, city
FROM employees;
```

Output

```
88 •  SELECT DISTINCT department, city
89    FROM employees;
90
91
```

esult Grid | Filter Rows:

department	city
IT	San Francisco
HR	Boston
Sales	Chicago
Marketing	New York
Finance	Boston

Part 2: Expressions in SELECT

Exercise 2.1: Arithmetic Operations

Task: Calculate the monthly salary for each employee (annual salary divided by 12).

Expected Result: 15 rows showing both annual and monthly salary

Your Query:

```
SELECT
    first_name,
    last_name,
    salary AS annual_salary,
    salary / 12 AS monthly_salary
FROM employees;
```

Output

```
90 •  SELECT
91        first_name,
92        last_name,
93        salary AS annual_salary,
94        salary / 12 AS monthly_salary
95    FROM employees;
96
97
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	annual_salary	monthly_salary
John	Watson	90000.00	7500.000000
Jane	Wilson	70000.00	5833.333333
Mark	Johnson	55000.00	4583.333333
Sara	Anderson	65000.00	5416.666667
Paul	Brown	85000.00	7083.333333
Emily	Clark	75000.00	6250.000000
David	Miller	60000.00	5000.000000
Anna	Taylor	78000.00	6500.000000
Robert	Wilson	52000.00	4333.333333
James	Thomas	54000.00	4500.000000
Linda	Martinez	72000.00	6000.000000
Kevin	White	58000.00	4833.333333
Rachel	Green	60000.00	5000.000000
Brian	Adams	68000.00	5666.666667
Nancy	Scott	56000.00	4666.666667

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Exercise 2.2: Multiple Calculations

****Task****: Display employee names with their annual salary, monthly salary, and weekly salary (annual/52).

****Expected Result****: 15 rows with salary broken down by different time periods

****Your Query****:

SELECT

first_name,

last_name,

salary AS annual_salary,

salary / 12 AS monthly_salary,

salary / 52 AS weekly_salary

FROM employees;

****Output****

```sql
SELECT
    first_name,
    last_name,
    salary AS annual_salary,
    salary / 12 AS monthly_salary,
    salary / 52 AS weekly_salary
FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	annual_salary	monthly_salary	weekly_salary
John	Watson	90000.00	7500.000000	1730.769231
Jane	Wilson	70000.00	5833.333333	1346.153846
Mark	Johnson	55000.00	4583.333333	1057.692308
Sara	Anderson	65000.00	5416.666667	1250.000000
Paul	Brown	85000.00	7083.333333	1634.615385
Emily	Clark	75000.00	6250.000000	1442.307692
David	Miller	60000.00	5000.000000	1153.846154
Anna	Taylor	78000.00	6500.000000	1500.000000
Robert	Wilson	52000.00	4333.333333	1000.000000
James	Thomas	54000.00	4500.000000	1038.461538
Linda	Martinez	72000.00	6000.000000	1384.615385
Kevin	White	58000.00	4833.333333	1115.384615
Rachel	Green	60000.00	5000.000000	1153.846154
Brian	Adams	68000.00	5666.666667	1307.692308
Nancy	Scott	56000.00	4666.666667	1076.923077

Exercise 2.3: String Concatenation

****Task****: Create a full name by combining first name and last name with a space in between.

****Hint****: Use the appropriate concatenation method for your database (|| for PostgreSQL/Oracle, CONCAT for MySQL)

****Expected Result****: 15 rows with combined full names

****Your Query****:

```
SELECT
  CONCAT(first_name, ' ', last_name) AS full_name
FROM employees;
```

****Output****

```
SELECT
    CONCAT(first_name, ' ', last_name) AS full_name
FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name
John Watson
Jane Wilson
Mark Johnson
Sara Anderson
Paul Brown
Emily Clark
David Miller
Anna Taylor
Robert Wilson
James Thomas
Linda Martinez
Kevin White
Rachel Green
Brian Adams
Nancy Scott

Part 3: WHERE Clause - Basic Filtering

Exercise 3.1: Equality Condition

Task: Find all employees who work in the IT department.

Expected Result: 5 employees

Your Query:

SELECT *

FROM employees

WHERE department = 'IT';

Output

```
SELECT *
FROM employees
WHERE department = 'IT';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 3.2: Numeric Comparison

Task: Find all employees with a salary greater than 70000.

Expected Result: 8 employees

Your Query:

SELECT *

FROM employees

WHERE salary > 70000;

Output

```
110 •  SELECT *
111    FROM employees
112    WHERE salary > 70000;
113
114
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 3.3: Not Equal To

Task: Find all employees who are NOT in the Sales department.

Expected Result: 10 employees

Your Query:

SELECT *FROM employees WHERE department <> 'Sales';

Output

```
113 •  SELECT *
114    FROM employees
115    WHERE department <> 'Sales';
116
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

employees 15 ×

Exercise 3.4: Less Than or Equal To

Task: Find all employees with a salary of 60000 or less.

Expected Result: 7 employees

Your Query:

```
SELECT *
FROM employees
WHERE salary <= 60000;
```

Output

```
116 • SELECT *
117   FROM employees
118   WHERE salary <= 60000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 4: Logical Operators

Exercise 4.1: AND Operator

Task: Find all employees in the IT department who earn more than 85000.

Expected Result: 3 employees

Your Query:

```
SELECT *
FROM employees
WHERE department = 'IT'
  AND salary > 85000;
```

Output

```
119 • SELECT *
120   FROM employees
121   WHERE department = 'IT'
122     AND salary > 85000;
123
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.2: OR Operator

Task: Find all employees who work in either Sales or Marketing departments.

Expected Result: 8 employees

Your Query:

SELECT *

FROM employees

WHERE department = 'Sales'

  OR department = 'Marketing';

Output

```
123 •  SELECT *
124    FROM employees
125    WHERE department = 'Sales'
126       OR department = 'Marketing';
127
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.3: Combining AND and OR

Task: Find employees in Sales or Marketing departments who earn more than 60000.

Hint: Use parentheses to group the OR conditions

Expected Result: 2 employees

Your Query:

SELECT *

FROM employees

WHERE (department = 'Sales' OR department = 'Marketing')

  AND salary > 60000;

Output

```
127 •  SELECT *
128    FROM employees
129    WHERE (department = 'Sales' OR department = 'Marketing')
130      AND salary > 60000;
131
132
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.4: NOT Operator

Task: Find all employees who are NOT in IT or HR departments.

Expected Result: 8 employees

Your Query:

SELECT *

FROM employees

WHERE department NOT IN ('IT', 'HR');

Output

```
SELECT *
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 5: Special WHERE Conditions

Exercise 5.1: BETWEEN Operator

Task: Find all employees with salaries between 55000 and 75000 (inclusive).

Expected Result: 9 employees

Your Query:

SELECT *

FROM employees

WHERE salary BETWEEN 55000 AND 75000;

```
SELECT *
FROM employees
WHERE salary BETWEEN 55000 AND 75000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.2: IN Operator

Task: Find all employees located in New York, Chicago, or Boston.

Expected Result: 9 employees

Your Query:

SELECT *

FROM employees

WHERE city IN ('New York', 'Chicago', 'Boston');

Output

```
SELECT *
FROM employees
WHERE city IN ('New York', 'Chicago', 'Boston');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.3: LIKE with % Wildcard

Task: Find all employees whose first name starts with 'J'.

Expected Result: 4 employees

Your Query:

SELECT *

FROM employees

WHERE first_name LIKE 'J%';

```
SELECT *
FROM employees
WHERE first_name LIKE 'J%';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francis
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.4: LIKE with Multiple Patterns

Task: Find all employees whose last name ends with 'son'.

Expected Result: 3 employees

Your Query:

SELECT *

FROM employees

WHERE last_name LIKE '%son';

Output

```
142 FROM employees
143 WHERE last_name LIKE '%son';
144
145
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.5: LIKE with _ Wildcard

Task: Find all employees whose first name is exactly 4 letters long.

Hint: Use four underscore characters

Expected Result: 3 employees

Your Query:

SELECT *

FROM employees

WHERE first_name LIKE '____';

```
145 SELECT *
146 FROM employees
147 WHERE first_name LIKE '____';
148
149
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.6: IS NULL

Task: Find all employees who have no manager (manager_id is NULL).

Hint: Do not use = with NULL

Expected Result: 3 employees

Your Query:

SELECT *

FROM employees

WHERE manager_id IS NULL;

Output

```
149 •  SELECT *
150    FROM employees
151    WHERE manager_id IS NULL;
152
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.7: IS NOT NULL

Task: Find all employees who have a manager.

Expected Result: 12 employees

Your Query:

SELECT *

FROM employees

WHERE manager_id IS NOT NULL;

```
152 •  SELECT *
153    FROM employees
154    WHERE manager_id IS NOT NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
14	Brian	Adams	brian.adams@gmail.com	Finance	Analyst	68000.00	2019-10-23	2	Boston
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

employees 27 ×

Part 6: Working with Products Table

Exercise 6.1: Low Stock Alert

Task: Find all products with stock quantity less than 50.

Expected Result: 8 products

Your Query:

SELECT *

FROM products

WHERE stock_quantity < 50;

Output

```
153 •  SELECT *
154    FROM products
155    WHERE stock_quantity < 50;
156
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	supplier
1	Laptop	Electronics	800.00	20	TechCorp
2	Smartphone	Electronics	600.00	35	TechCorp
4	Keyboard	Accessories	45.00	40	TechCorp
6	Monitor	Electronics	180.00	0	DisplayHub
8	Webcam	Electronics	90.00	25	VisionTech
9	Power Bank	Electronics	60.00	45	ChargePlus
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 6.2: Out of Stock

Task: Find all products that are completely out of stock (stock quantity equals 0).

Expected Result: 1 product

Your Query:

SELECT *

FROM products

WHERE stock_quantity = 0;

Output

```
154 •  SELECT *
155    FROM products
156    WHERE stock_quantity = 0;
157
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	supplier
6	Monitor	Electronics	180.00	0	DisplayHub
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 6.3: Price Range

Task: Find all products priced between 20 and 100 dollars.

Expected Result: 7 products

Your Query:

SELECT *

FROM products

WHERE price BETWEEN 20 AND 100;

Output

```
155 •   SELECT *
156     FROM products
157     WHERE price BETWEEN 20 AND 100;
158
159
```

Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	product_name	category	price	stock_quantity	supplier
3	Headphones	Accessories	50.00	120	SoundWave
4	Keyboard	Accessories	45.00	40	TechCorp
5	Mouse	Accessories	25.00	150	CableWorks
8	Webcam	Electronics	90.00	25	VisionTech
9	Power Bank	Electronics	60.00	45	ChargePlus
10	Laptop Bag	Accessories	35.00	60	CarryAll
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 6.4: Specific Categories

Task: Find all products in Electronics or Accessories categories.

Expected Result: 13 products

Your Query:

SELECT *

FROM products

WHERE category IN ('Electronics', 'Accessories');

```
158 •   SELECT *
159     FROM products
160     WHERE category IN ('Electronics', 'Accessories');
161
```

Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	product_name	category	price	stock_quantity	supplier
1	Laptop	Electronics	800.00	20	TechCorp
2	Smartphone	Electronics	600.00	35	TechCorp
3	Headphones	Accessories	50.00	120	SoundWave
4	Keyboard	Accessories	45.00	40	TechCorp
5	Mouse	Accessories	25.00	150	CableWorks
6	Monitor	Electronics	180.00	0	DisplayHub
7	USB Cable	Accessories	10.00	300	CableWorks
8	Webcam	Electronics	90.00	25	VisionTech
9	Power Bank	Electronics	60.00	45	ChargePlus
10	Laptop Bag	Accessories	35.00	60	CarryAll
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 6.5: Supplier Search

Task: Find all products supplied by TechCorp.

Expected Result: 6 products

Your Query:

SELECT *

FROM products

WHERE supplier = 'TechCorp';

Output

```
159 •  SELECT *
160    FROM products
161    WHERE supplier = 'TechCorp';
162
```

Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	product_name	category	price	stock_quantity	supplier
1	Laptop	Electronics	800.00	20	TechCorp
2	Smartphone	Electronics	600.00	35	TechCorp
4	Keyboard	Accessories	45.00	40	TechCorp
NULL	NULL	NULL	NULL	NULL	NULL

Part 7: Working with Orders Table

Exercise 7.1: Recent Orders

Task: Find all orders placed in March 2024.

Hint: Use BETWEEN with dates in 'YYYY-MM-DD' format

Expected Result: 4 orders

Your Query:

SELECT *FROM orders WHERE order_date BETWEEN '2024-03-01' AND '2024-03-31';

Output

```
162 •  SELECT *
163    FROM orders
164    WHERE order_date BETWEEN '2024-03-01' AND '2024-03-31';
165
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

order_id	customer_name	product_id	quantity	order_date	total_amount	status
1	Alice	1	1	2024-03-10	800.00	Shipped
4	David	6	1	2024-03-22	180.00	Processing
6	Frank	8	2	2024-03-05	180.00	Shipped
8	Helen	9	2	2024-03-14	120.00	Processing
10	Jack	7	5	2024-03-19	50.00	Shipped
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.2: High Value Orders

Task: Find all orders with a total amount greater than 200.

Expected Result: 8 orders

Your Query:

SELECT *

FROM orders

WHERE total_amount > 200;

Output

```
163 •  SELECT *
164    FROM orders
165    WHERE total_amount > 200;
166
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

order_id	customer_name	product_id	quantity	order_date	total_amount	status
1	Alice	1	1	2024-03-10	800.00	Shipped
5	Emma	2	1	2024-02-28	600.00	Delivered
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.3: Order Status

Task: Find all orders that are currently being processed or shipped.

Expected Result: 4 orders

Your Query:

SELECT * FROM orders WHERE status IN ('Processing', 'Shipped');

Output

```
166 •  SELECT *
167    FROM orders
168    WHERE status IN ('Processing', 'Shipped');
169
```

Result Grid | Filter Rows: | Edit: | Export/Import:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
1	Alice	1	1	2024-03-10	800.00	Shipped
4	David	6	1	2024-03-22	180.00	Processing
6	Frank	8	2	2024-03-05	180.00	Shipped
8	Helen	9	2	2024-03-14	120.00	Processing
10	Jack	7	5	2024-03-19	50.00	Shipped
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.4: Cancelled Orders

Task: Find all cancelled orders.

Expected Result: 1 order

Your Query:

```
SELECT *
FROM orders
WHERE status = 'Cancelled';
```

Output

```
SELECT *
FROM orders
WHERE status = 'Cancelled';
```

order_id	customer_name	product_id	quantity	order_date	total_amount	status
3	Chris	4	2	2024-01-05	90.00	Cancelled
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 8: Complex Queries

Exercise 8.1: Department Analysis

Task: Find all IT employees in San Francisco who earn more than 80000.

Expected Result: 3 employees

Your Query:

```
SELECT *
FROM employees
WHERE department = 'IT'
  AND city = 'San Francisco'
  AND salary > 80000;
```

Output

```
FROM employees
WHERE department = 'IT'
  AND city = 'San Francisco'
  AND salary > 80000;
```

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.2: Hiring Date Range

Task: Find all employees hired between January 1, 2020 and December 31, 2021.

Expected Result: 9 employees

Your Query:

SELECT *

FROM employees

WHERE hire_date BETWEEN '2020-01-01' AND '2021-12-31';

Output

```
175 •  SELECT *
176    FROM employees
177    WHERE hire_date BETWEEN '2020-01-01' AND '2021-12-31';
178
179
180
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
11	Linda	Martinez	linda.martinez@gmail.com	IT	QA Engineer	72000.00	2021-04-12	1	San Francisco
13	Rachel	Green	rachel.green@gmail.com	Marketing	SEO Specialist	60000.00	2020-08-19	4	New York
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.3: Email Domain Search

Task: Find all employees whose email contains 'company.com'.

Expected Result: 15 employees (all employees)

Your Query:

SELECT * FROM employeesWHERE email LIKE '%company.com%';

Output

```
179 •  SELECT *
180    FROM employees
181    WHERE email LIKE '%company.com%';
182
183
184
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
5	Paul	Brown	paul.brown@company.com	IT	System Engineer	85000.00	2018-11-11	1	San Francisco
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
7	David	Miller	david.miller@company.com	Finance	Accountant	60000.00	2021-07-01	2	Boston
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.4: Product Inventory Alert

Task: Find all Electronics products that either have zero stock or less than 30 units, and price above 50.

Hint: Use parentheses to group the OR condition for stock

Expected Result: 2 products

Your Query:

```
SELECT * FROM products WHERE category = 'Electronics'
  AND (stock_quantity = 0 OR stock_quantity < 30)
  AND price > 50;
```

Output

```
183 •  SELECT *
184    FROM products
185    WHERE category = 'Electronics'
186      AND (stock_quantity = 0 OR stock_quantity < 30)
187      AND price > 50;
188
189
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	supplier
1	Laptop	Electronics	800.00	20	TechCorp
6	Monitor	Electronics	180.00	0	DisplayHub
8	Webcam	Electronics	90.00	25	VisionTech
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.5: Multi-Condition Order Search

Task: Find all delivered orders from February or March 2024 with total amount over 150.

Expected Result: 2 orders

Your Query:

```
SELECT *
FROM orders
WHERE status = 'Delivered'
  AND order_date BETWEEN '2024-02-01' AND '2024-03-31'
  AND total_amount > 150;
```

```
188 •  SELECT *
189    FROM orders
190    WHERE status = 'Delivered'
191      AND order_date BETWEEN '2024-02-01' AND '2024-03-31'
192      AND total_amount > 150;
193
194
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
5	Emma	2	1	2024-02-28	600.00	Delivered
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 9: Challenge Exercises

Challenge 9.1: Salary Band Analysis

Task: Categorize employees into salary bands. Find all employees earning between 50000-60000 and label them as 'Entry Level'.

Hint: Use a string literal in SELECT to add the label

Expected Result: Employees in entry level salary band

Your Query:

```
SELECT
    employee_id,
    first_name,
    last_name,
    salary,
    'Entry Level' AS salary_band
FROM employees
WHERE salary BETWEEN 50000 AND 60000;
```

Output

```
193 •  SELECT
194        employee_id,
195        first_name,
196        last_name,
197        salary,
198        'Entry Level' AS salary_band
199    FROM employees
200    WHERE salary BETWEEN 50000 AND 60000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_id	first_name	last_name	salary	salary_band
3	Mark	Johnson	55000.00	Entry Level
7	David	Miller	60000.00	Entry Level
9	Robert	Wilson	52000.00	Entry Level
10	James	Thomas	54000.00	Entry Level
12	Kevin	White	58000.00	Entry Level
13	Rachel	Green	60000.00	Entry Level
15	Nancy	Scott	56000.00	Entry Level

Challenge 9.2: Name Pattern Search

Task: Find employees whose first name contains 'ar' OR last name starts with 'W' OR 'T'

Expected Result: 7 employees

Your Query:

SELECT *

FROM employees

WHERE first_name LIKE '%ar%'

  OR last_name LIKE 'W%'

  OR last_name LIKE 'T%';

Output

```
201 •  SELECT *
202    FROM employees
203    WHERE first_name LIKE '%ar%'
204       OR last_name LIKE 'W%'
205       OR last_name LIKE 'T%';
206
207
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Watson	john.watson@company.com	IT	Senior Developer	90000.00	2020-03-15	NULL	San Francisco
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
4	Sara	Anderson	sara.anderson@company.com	Marketing	Marketing Lead	65000.00	2020-09-05	2	New York
8	Anna	Taylor	anna.taylor@company.com	IT	Developer	78000.00	2020-05-22	1	San Francisco
9	Robert	Wilson	robert.wilson@company.com	Marketing	Content Specialist	52000.00	2022-03-14	4	New York
10	James	Thomas	james.thomas@company.com	HR	Recruiter	54000.00	2021-11-30	2	Boston
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.3: Product Reorder List

Task: Create a reorder list for products that are:

- In stock (quantity > 0)

- Low stock (quantity < 50)

- Price under 100

- From specific suppliers (TechCorp or CableWorks)

Expected Result: Products matching all criteria

Your Query:

SELECT *

FROM products

WHERE stock_quantity > 0

  AND stock_quantity < 50

  AND price < 100

  AND supplier IN ('TechCorp', 'CableWorks');

Output

```
SELECT *
FROM products
WHERE stock_quantity > 0
  AND stock_quantity < 50
  AND price < 100
  AND supplier IN ('TechCorp', 'CableWorks');
```

Result Grid | Filter Rows: | Edit: | Export/Imp

product_id	product_name	category	price	stock_quantity	supplier
4	Keyboard	Accessories	45.00	40	TechCorp
NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.4: Employee Location and Department

Task: Find employees who:

- Work in Sales, Marketing, or HR

- Are located in cities that start with 'C' or 'B'

- Have a salary not equal to 54000 or 52000

Expected Result: 4 employees

Your Query:

```
SELECT *
FROM employees
WHERE department IN ('Sales', 'Marketing', 'HR')
  AND (city LIKE 'C%' OR city LIKE 'B%')
  AND salary NOT IN (54000, 52000);
```

Output

```
SELECT *
FROM employees
WHERE department IN ('Sales', 'Marketing', 'HR')
  AND (city LIKE 'C%' OR city LIKE 'B%')
  AND salary NOT IN (54000, 52000);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Wilson	jane.wilson@company.com	HR	HR Manager	70000.00	2019-06-10	1	Boston
3	Mark	Johnson	mark.johnson@company.com	Sales	Sales Executive	55000.00	2021-01-20	2	Chicago
6	Emily	Clark	emily.clark@company.com	Sales	Sales Manager	75000.00	2019-02-18	1	Chicago
12	Kevin	White	kevin.white@gmail.com	Sales	Sales Associate	58000.00	2022-06-01	3	Chicago
15	Nancy	Scott	nancy.scott@gmail.com	HR	HR Executive	56000.00	2021-12-05	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.5: Order Investigation

Task: Find orders where:

- Customer name starts with a letter between 'A' and 'H'

- Order date is NOT in January 2024

- Status is NOT cancelled

- Quantity ordered is between 2 and 6

Hint: Use string comparison for customer name filter

Expected Result: 5 orders

Your Query:

```
SELECT *
FROM orders
WHERE customer_name BETWEEN 'A' AND 'H'
  AND order_date NOT BETWEEN '2024-01-01' AND '2024-01-31'
  AND status <> 'Cancelled'
  AND quantity BETWEEN 2 AND 6;
```

Output

```
217  SELECT *
218  FROM orders
219  WHERE customer_name BETWEEN 'A' AND 'H'
220    AND order_date NOT BETWEEN '2024-01-01' AND '2024-01-31'
221    AND status <> 'Cancelled'
222    AND quantity BETWEEN 2 AND 6;
223
```

Result Grid | Filter Rows: | Edit: | Export/Import:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
2	Bob	5	3	2024-02-15	75.00	Delivered
6	Frank	8	2	2024-03-05	180.00	Shipped
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Self-Assessment Checklist

After completing the lab, check off each skill you feel confident with:

- [ ] Writing basic SELECT statements

- [ ] Selecting specific columns

- [ ] Using column aliases

- [ ] Using SELECT DISTINCT

- [ ] Performing arithmetic operations in SELECT
- [ ] Concatenating strings
- [ ] Using WHERE clause with comparison operators
- [ ] Using AND, OR, NOT logical operators
- [ ] Using BETWEEN operator
- [ ] Using IN operator
- [ ] Using LIKE operator with wildcards
- [ ] Using IS NULL and IS NOT NULL
- [ ] Combining multiple conditions with proper parentheses
- [ ] Writing complex queries with multiple tables
- [ ] Troubleshooting SQL syntax errors

Tips for Success

1. **Test incrementally**: Start with simple queries and add complexity gradually
2. **Check row counts**: Verify your results match the expected number of rows
3. **Use proper formatting**: Indent your queries for better readability
4. **Remember quotes**: Text values need single quotes, numbers don't
5. **NULL is special**: Use IS NULL, not = NULL
6. **Parentheses matter**: Use them to control logic in complex conditions
7. **LIKE is case-sensitive**: Depending on your database configuration
8. **Ask questions**: If you're stuck, ask your instructor for guidance

Common Mistakes to Avoid

1. Forgetting quotes around text values
2. Using = instead of IS NULL
3. Missing parentheses in complex conditions
4. Incorrect wildcard usage (% vs _)
5. Wrong date format
6. Confusing AND/OR precedence

Notes Section

Use this space to write down important points, questions, or observations: